AgriEYE 360: cotton field pipeline

AgriEYE 360 turns 360° photos of a cotton plot into results for each plant:

- open bolls and closed bolls
- leaf health (healthy or diseased leaves, % diseased)
- growth from one visit to the next
- a Street View style tour of the field

Last updated: 9 Nov 2026

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1. Status and results

Stage	Status	Result
Import 360 photos	● Done	35 unique panoramas from 6 data folders (765 files; the rest were copies). 5 visits: 22 Jul (1 photo, no plant ID), 31 Jul, 4 Aug, 6 Aug, 20 Aug 2026
Plant IDs (stations)	● Done, checked 6 Nov	14 plants (P01–P13). Repeat shots removed from the analysis. At most 1 photo per plant per visit (30 photos used)
Slice into 4 views (N/E/S/W)	● Done	1440 px, 90° views, north-aligned. Analysis: 120 views. Training data: 140 views (all 35 photos)
Leaf crops (SAM 3) + manual sorting	● Done	diseased 1,259 · healthy 5,424 · rejected 7,231
Disease classifier v2 (YOLO11n-cls)	● Trained 6 Nov	89.2% accuracy on 835 field test leaves, ROC AUC 0.944. Only 22% on outside close-up photos (does not generalise)
Leaf finder (YOLO11n-seg)	● Trained 6 Nov	Mask precision 0.69, recall 0.62, mAP50 0.67, mAP50-95 0.40
SAM 3 leaf segmentation (text prompt "leaf")	● Done	5,085 leaves in 120 views, median score 0.57
SAM 3 leaf outlines, hand check	● Done 9 Nov	426 outlines on 10 views: **all real leaves** (precision 100%), 423 with a good fit (99.3%). Recall not measured (see 1e)
SAM 3 + YOLO v2 disease, whole field	● Done	4,231 leaves judged. % diseased: 31 Jul 15.3 · 4 Aug 21.4 · 6 Aug 13.2 · 20 Aug 25.6
Boll labels: SAM 3 + hand sorting	● Done 8 Nov	3,068 boll boxes; 2,970 hand-checked: open 441 · closed 2,465 · not a boll 64
Boll classifier (Roboflow ViT-Base)	● Trained 8 Nov	**97.7% on 355 test images** (open recall 91.7%, closed 100%, sky/cloud rejected 51/51)
Boll detector (Roboflow, one step), comparison only	● Trained 9 Nov	**About 41% mAP@50** (about 21% mAP@50:95) after about 320 epochs. Not used in the analysis (see 1f)
4-POV training datasets for Roboflow	● Done 8 Nov	5 datasets in `roboflow_export\` (see 1g)
Field analysis, report, Street View	● Done 8 Nov	Re-run with new plant IDs, leaf finder, disease v2 and boll labels (hand > Roboflow model > SAM 3)
Disease classifier v3 (field + public data)	● Data ready, not trained	`train_disease_v3.py train` (only when you decide to train)

1a. Field results (growth.csv)

Open bolls per plant (sum of the 4 views):

Plant	31 Jul	4 Aug	6 Aug	20 Aug
P02	0	3	13	27
P03	0	11	–	53
P06	0	6	–	53
P09	2	2	5	41
P10	–	3	1	66
P13	–	–	15	74

- **Bolls are opening, not disappearing:** closed bolls fall at the same time (P06 110 → 18, P02 71 → 24, P09 105 → 26).
- **Leaf disease rises by 20 Aug:** P09 34%, P13 31%, P03 29%, P06 26% diseased leaves (31 Jul: about 7–25%).

- **Boll labels used:** 2,557 from your hand sorting, 85 from the Roboflow model; 9 uncertain model answers (`check_bolls`).
- **Caveat:** the 4 Aug photos were taken in evening light, which makes red/orange leaves look diseased; compare 31 Jul, 6 Aug and 20 Aug first.

1b. Disease classifier v2: performance

Tested on 835 leaves from photos the model never saw during training. Every visit is included in the test set.

	Precision	Recall
Diseased	85.7%	82.7%
Healthy	91.0%	92.7%

- Confusion matrix: 239 diseased found, 50 missed, 40 healthy wrongly flagged, 506 healthy correct.
- Accuracy by visit: 31 Jul 92% · 4 Aug (evening light) 85% · 6 Aug 94% · 20 Aug 88%.
- **"Needs checking" band:** leaves with a diseased probability between 0.2 and 0.8 get a yellow outline and are counted in `check_leaves`. The model is confident on 74% of leaves, and 94% of those are correct.
- **Can SAM 3 judge disease by itself?** No. The prompt "diseased leaf" picks almost every leaf (precision 43%, AUC 0.45). The best SAM 3 rule reached 64% accuracy (AUC 0.75) versus 89% (AUC 0.944) for v2. So SAM 3 is used to **find** leaves and the trained classifier **judges** them.
- **Outside photos:** v2 scores 22% on public close-up leaf photos. Public CC BY 4.0 datasets (Mendeley t9hgvk2h9p and SAR-CLD-2024 b3jy2p6k8w) are prepared for v3 (2,681 leaf crops).

1c. Boll classifier (Roboflow) and the cloud problem

- **Pipeline:** SAM 3 finds every boll (prompts "open cotton boll", "closed cotton boll", "flower"), the boll classifier decides open / closed / not a boll.
- **Why not SAM 3's own label:** of the boxes SAM 3 called "open", only about half were open; "flower" was almost never a flower (red closed bolls of this red-leaf variety).
- **Hand sorting:** `sort_bolls_grid.py` shows 30 crops per page with SAM 3's guess filled in; you only click the wrong ones. Your sorting changed SAM 3's answer on 339 boxes.
- **Cloud vs cotton:** an early Roboflow model mixed up white clouds and open cotton. Fix: 383 sky/cloud patches (smooth, away from any boll) added as `not_boll`, plus no brightness augmentation.
- **Model:** Roboflow ViT-Base (ImageNet-21k), 224×224, flip + ±15° rotation, 3×, early stopping (0.16 credits). Roboflow id `cotton-boll-open-or-close-3frah/3`.
- **Test (355 images never used in training, `eval_roboflow_boll.py`):**

Class	Precision	Recall	Test images
open_boll	95.7%	91.7%	48
closed_boll	97.7%	100%	252
not_boll	100%	92.7%	55

- The 8 mistakes are bolls just starting to crack, tiny crops and one cloud-filled crop; 3 had low confidence (0.43–0.56). Answers below 0.6 are counted as `check_bolls`.
- **Limitation:** only 4 test images were real non-sky "not a boll" objects (all 4 wrong); open bolls are rare (48 in test).
- On all 3,068 field crops the model agrees with your hand labels 95.4% (includes training crops, so this is a consistency check, not a test score).

1d. Which models are used (final choice)

Every task is done in **two steps**: SAM 3 finds the objects, then a trained classifier judges each one.

Task	Step 1: find	Step 2: judge	Score	Trained where
Cotton bolls	SAM 3 (ready-made, not trained)	Boll classifier, open / closed / not a boll	97.7%	Roboflow website

Task	Step 1: find	Step 2: judge	Score	Trained where
Leaves and disease	SAM 3 / leaf finder (YOLO11n-seg)	Disease classifier v2, healthy / diseased	89.2%	Local machine

- Roboflow models: boll classifier (used) and boll detector (comparison only, see 1f).
- Local models: leaf finder `disease_autolabel\models\leaf_seg\weights\best.pt` and disease v2 `disease_autolabel\models\disease_cls\weights\best.pt`. `disease_cls_v1_backup\` is the older v1 of the same model. Disease v3 is not trained.

1e. SAM 3 leaf outlines: hand check

`sam3_seg_check.py` showed 10 views (spread over all 4 visits) with every SAM 3 leaf outline (prompt "leaf", conf 0.3).

- **Precision: 426 of 426 outlines (100%) were judged to be real leaves.** 423 had a good fit (99.3%); 3 were partial (all in P09, 31 Jul, east view: small dead leaves on the soil). No boll, stem, sky or soil was outlined. A few outlines are weed or grass leaves, not cotton.
- **Two rounds:** a precision round (`answers.json`) and a recall round (`answers_recall.json`, filled outlines, missed leaves to be marked). The report uses the recall round.
- **Recall: not measured.** Leaves with no outline were not marked. A visual check shows SAM 3 **misses some clearly visible leaves**, so recall is below 100%.
- **What this means:** SAM 3 gives a reliable **sample** of leaves on each plant. That is fine for **% diseased** (a ratio). **Leaf counts are not used as a measure of plant size.**
- Files: `F:\Project and completed courses\Agri EYE 360\sample_results\sam3_seg_check\` (`report.txt`, `per_view.csv`, `answers.json`).

1f. One-step boll detector (Roboflow): comparison

- **Data:** `1_cotton_boll_detection` (140 views, about 3,000 boll boxes: open 453, closed 2,544; split 94 / 28 / 18 views), project "Cotton boll labelling".
- **Result (Roboflow training graphs):** mAP@50 rose to about 35% by epoch 40 and ended at **about 41%** after about 320 epochs; mAP@50:95 about 21%. The losses were flat after about 30 epochs (no overfitting; more epochs do not help).
- **Why it is lower:** most bolls are only 20–40 px in a 1440 px view (smaller still at 640 px input); closed bolls are green on green leaves; only 140 photos (28 for validation, so the curve is noisy); the labels come from SAM 3 boxes, so real bolls that SAM 3 missed count as false detections.
- **Conclusion:** with this little data, **two steps (SAM 3 + classifier, 97.7%) work much better than one detector**. The analysis uses the two-step method.

1g. Roboflow datasets (all built from the 4-POV views)

Folder: `F:\Project and completed courses\Agri EYE 360\roboflow_export\`

Zip	Roboflow project type	Contents
`1_cotton_boll_detection.zip`	Object Detection	140 views, 3,004 boll boxes (2,906 hand-checked); split 94 / 28 / 18 views
`2_cotton_boll_classification.zip`	Classification (single-label)	open 441, closed 2,465, not_boll 64 + 383 sky; split train 2,328 / valid 670 / test 355
`2_cotton_boll_classification_csv.zip`	Classification	Same data, labels in `_classes.csv` (fallback if folder labels are not read)
`3_leaf_segmentation.zip`	Instance Segmentation	140 views, about 11,200 leaf outlines
`4_leaf_disease_field.zip`	Classification	Field leaves cut from the views: diseased 816, healthy 2,263
`5_leaf_disease_field_outside.zip`	Classification	Dataset 4 + 2,681 public leaves

- **Labels moved from panorama to views:** the hand-labelled leaf outlines were projected with the exact inverse of the slicer's maths (checked visually). 10,108 leaves landed on a view; 3,785 were above or below the 45°

view limit.

- **Split without leakage:** grouped by plant + visit (+ view direction for the boll sets; the 4 views do not overlap). Repeat shots always stay with their twin. The boll sets are balanced so open bolls are also 70 / 20 / 10.
- **Roboflow tips learned:** choose "Use Existing Values" for the split (never the random slider); Roboflow needs a test split of at least 2 images; the hosted API needs the **private** API key (the `rf_…` publishable key gives 401).

1h. Plant ID fixes (6 Nov)

- **Repeat shots blanked**, keeping the sharper first shot: P04 6 Aug 12:35 · P07 4 Aug 17:26 · P06 4 Aug 17:19:47 · P10 6 Aug 12:44 (same image as 12:43).
- **P06 split in two.** P12 is the stop on the walk back: 31 Jul 08:05 and 4 Aug 17:13.
- **North-east end assigned by walking order:** P09 first stop, P10 second, P13 third.
- Photos cannot confirm plant identity across visits (feature matching scores about the same for same and different plants). The split relies on: two separate stops on one walk are two plants, and the walking order is the same every visit (confirmed). Treat P06/P12 and P09/P10/P13 as best guesses.

1i. Lessons learned

- **Label noise was the main problem** for disease: about 1/3 of the first "diseased" crops were clean leaves in evening light. A strict second review raised diseased recall from 54% to 83%.
- **Hold-key accident:** 662 crops got the same label from a held key; re-checked, and all sorters now ignore a held key.
- **Same photo, two timestamps:** `360 files data\2.jpg` and `6.jpg` are the same pictures as P02/P11 on 31 Jul with EXIF times 30 s apart. Duplicates are matched by image content and the copy with the plant ID is kept.
- **Training images should look like the images the model will see:** all training data now comes from the 4-POV views, not the stretched panoramas.
- **Clouds look like open cotton** at low resolution: add sky examples, avoid brightness augmentation, and tile detection images in Roboflow.
- **A hand check must also look for what is missing:** in the SAM 3 outline check every outline starts green, so "all green" only tells you the outlines are right, not that every leaf was found.
- **Small objects favour two steps:** a one-step boll detector reached about 41% mAP@50, while SAM 3 + a crop classifier reached 97.7%.
- **Panoramas:** the TOZO AnyVue 360 compass heading is correct (checked against the sun: east at 08:00, west at 17:00); GPS is only accurate to about 2–3 m.

2. Commands (Windows cmd)

Always use the venv Python. Open **cmd** and set two shortcuts first:

```
set PY="C:\Users\bhara\Downloads\.venv\Scripts\python.exe"
set P=F:\Project and completed courses\Agri EYE 360
```

2a. Field pipeline (`agrieye.py`)

One click: double-click `F:\Project and completed courses\Agri EYE 360\run_pipeline.bat` (import → stations → slice → autolabel → analyze → report → streetview, skips finished steps, log in `agrieye_output\pipeline_log.txt`).

```
%PY% "%P%\agrieye.py" import
%PY% "%P%\agrieye.py" stations
%PY% "%P%\check_stations.py"
%PY% "%P%\agrieye.py" slice
%PY% "%P%\agrieye.py" autolabel
%PY% "%P%\agrieye.py" analyze --use-sam3-labels
%PY% "%P%\agrieye.py" report
%PY% "%P%\agrieye.py" streetview
```

Command	What it does
`import`	Reads GPS, time and heading from the photos and writes `catalog.csv`
`stations`	Groups photos into plant IDs (`stations.csv`); your edits are kept unless you add `--force`
`check_stations.py`	Picture sheet and map for checking the plant IDs
`slice`	Cuts each 360 photo into N/E/S/W views
`autolabel`	SAM 3 boll/flower boxes on all views
`analyze`	Bolls (hand label > Roboflow model > SAM 3), leaves, % diseased, canopy, growth tables and plots
`report`	Annotated views and one summary sheet per plant
`streetview`	Tour JSON for the field viewer, one per visit

2b. Bolls: sorting, Roboflow datasets, Roboflow model

```
:: 1. SAM 3 boxes for the extra (training-only) views, then cut crops for sorting
%PY% "%P%\boll_extra_views.py"
%PY% "%P%\make_boll_crops.py"

:: 2. Hand sorting: grid of 30, click to fix, Enter = accept page, U = undo
%PY% "%P%\sort_bolls_grid.py"

:: 3. Roboflow datasets 1 + 2 (boll detection, boll classification with sky negatives)
%PY% "%P%\make_roboflow_datasets.py"

:: 4. Roboflow model: set your PRIVATE key in this cmd window first (never save it in a file)
set ROBOFLOW_API_KEY=your_private_key
%PY% "%P%\eval_roboflow_boll.py"
%PY% "%P%\boll_classify_roboflow.py"
```

- `eval_roboflow_boll.py` tests the model on the 355 test images (`--limit 5` for a quick check) → `sample_results\roboflow_boll_test\`.
- `boll_classify_roboflow.py` classifies every SAM 3 boll box (about 0.2 credits, resumable) → `agrieye_output\objects\boll_cls_roboflow.csv`, used by `analyze`.

2c. 4-POV training datasets (all models)

```
%PY% "%P%\make_pov_dataset.py" all
```

Finds all unique 360 photos in every Data set folder, slices them, moves the hand leaf labels onto the views and writes Roboflow datasets 3–5 (and calls `make_roboflow_datasets.py` for 1–2).

2d. Leaves and disease (`pipeline.py` and helpers)

```
:: Leaf crops from the panoramas with SAM 3, sorting, reviews
%PY% "%P%\sam3_autolabel.py"
%PY% "%P%\sort_crops.py"
%PY% "%P%\find_clean_diseased.py"
%PY% "%P%\sort_crops.py" --review diseased
%PY% "%P%\find_suspect_healthy.py" --n 500
%PY% "%P%\sort_crops.py" --review healthy

:: Disease classifier and leaf finder (training: only when you decide to train)
%PY% "%P%\build_disease_dataset.py" --max-healthy 2500
%PY% "%P%\pipeline.py" traincls
%PY% "%P%\pipeline.py" leafdata
%PY% "%P%\pipeline.py" trainleaf
%PY% "%P%\model_metrics.py"

:: SAM 3 leaf segmentation of the field, alone or + YOLO disease
%PY% "%P%\sam3_segment_field.py"
%PY% "%P%\sam3_segment_field.py" --classify

:: SAM 3 on any image (text prompt), disease-prompt experiment, model tests
%PY% "%P%\vlm_segment.py" "<image or folder>" --prompt "leaf"
%PY% "%P%\sam3_disease_test.py" report
%PY% "%P%\test_all_models.py"
%PY% "%P%\verify_leaves.py"

:: Disease v3 with public data (data built; train only when you decide)
%PY% "%P%\prepare_external.py"
%PY% "%P%\train_disease_v3.py" build
%PY% "%P%\train_disease_v3.py" train
%PY% "%P%\train_disease_v3.py" compare
```

2e. This guide

```
%PY% "%P%\make_readme_pdf.py"
```

Still open

1. Optional: SAM 3 recall. Run `sam3_seg_check.py` again (after renaming `answers.json`) and right-click every clear leaf with no outline.
2. Optional: tiled test of the Roboflow boll detector on the 18 test views (open and closed scored separately).
3. Optional: train the leaf disease model on Roboflow (`5_leaf_disease_field_outside.zip`) or v3 locally, and compare with v2.

SAM 3 hand check, after sorting:

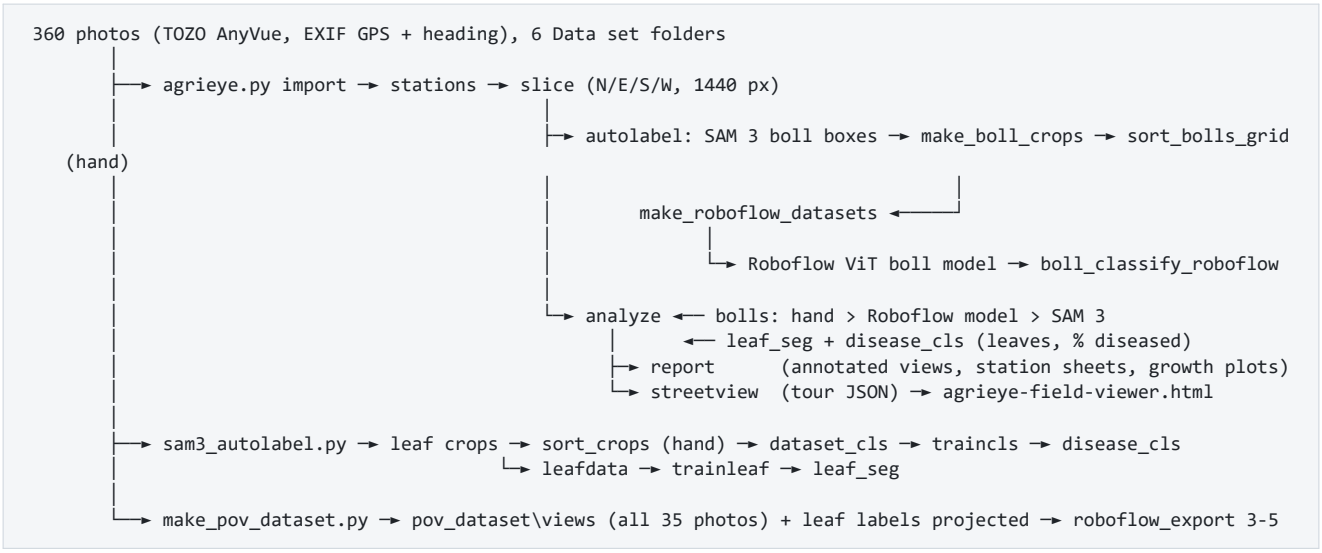
```
%PY% "%P%\sam3_seg_check.py"
%PY% "%P%\sam3_seg_check.py" --report
```

Clear open / closed boll pictures (cyan solid = open, magenta dashed = closed):

```
%PY% "%P%\make_boll_figures.py"
%PY% "%P%\make_boll_figures.py" --plant P13
```

3. Architecture

Data flow



Scripts (all in `F:\Project and completed courses\Agri EYE 360\`)

File	Purpose
`agrieye.py` + `agrieye_config.json`	Field pipeline: import, stations, slice, autolabel, refine, traindet, analyze, report, streetview, all
`run_pipeline.bat`	One-click field pipeline with log
`pipeline.py`	Leaf and disease pipeline: status, presort, leafdata, trainleaf, traincls, prelabel, predict
`sam3_autolabel.py`	SAM 3 leaf segmentation of the panoramas into crops and manifest
`sort_crops.py`, `find_clean_diseased.py`, `find_suspect_healthy.py`, `score_uncertain.py`	Leaf sorting and review tools
`build_disease_dataset.py`, `model_metrics.py`	Disease dataset and performance charts
`check_stations.py`	Plant-ID picture sheet, map and warnings
`make_boll_crops.py`, `sort_bolls.py`, `sort_bolls_grid.py`, `check_flowers.py`	Boll crops, sorters (one-by-one and grid), flower check sheets
`boll_extra_views.py`	SAM 3 boll boxes for the training-only views (22 Jul photo)
`make_pov_dataset.py`	All 35 photos → 4-POV views, leaf labels projected, Roboflow datasets 3–5
`make_roboflow_datasets.py`, `export_roboflow_bolls.py`	Roboflow datasets 1–2, 70/20/10 split, sky negatives
`eval_roboflow_boll.py`	Tests the Roboflow boll model on the test split
`boll_classify_roboflow.py`	Classifies every boll box with the Roboflow model for `analyze`
`sam3_segment_field.py`	SAM 3 leaf segmentation of the field (`--classify` adds YOLO disease)
`vlm_segment.py`	SAM 3 text-prompt segmentation of any image
`sam3_disease_test.py`	Experiment: can SAM 3 judge disease by itself (no)
`sam3_seg_check.py`	Hand check of SAM 3 leaf outlines (good / partial / wrong, missed leaves) → precision and recall
`make_boll_figures.py`	Open / closed boll pictures for the thesis (cyan solid / magenta dashed)
`prepare_external.py`, `train_disease_v3.py`	Public data and disease v3 (field + outside)
`test_all_models.py`, `verify_leaves.py`, `review_leaf_sorting.py`	Model tests and label checks
`make_readme_pdf.py`	Builds `AgriEYE360_project_guide.pdf` from this README

File	Purpose
`sam3.pt`	SAM 3 weights (3.4 GB; do not put in GitHub)
`AgriEYE st.Viwer\agrieye-field-viewer.html`	Street View viewer that loads the tour JSON

Input data

Location	Contents
`F:\Project and completed courses\Agri EYE 360\AgriEye 360 field data\Data set\overall dataset\`	The 34 panoramas used for the analysis
`...\Data set\360 files data\`, `fourth iteration\`, `Second 100 data sets\`, `Third new data set\`	Other copies; add the 22 Jul photo for training
`F:\Project and completed courses\Agri EYE 360\external_data\`	Public CC BY 4.0 leaf datasets (downloads, extracted, crops)

`F:\Project and completed courses\Agri EYE 360\agrieye_output\` (field results)

Location	Contents
`...\agrieye_output\catalog.csv`, `stations.csv`, `stations_backup_auto.csv`	Photos, **plant ID for each photo** (edit `station`), automatic IDs before the fix
`...\agrieye_output\station_check\`	Plant-ID sheet and GPS map
`...\agrieye_output\povs\P01\` … `P13\`, `_skipped\`	4 views per photo used in the analysis; views of blanked repeat shots
`...\agrieye_output\objects\sam3_labels.jsonl`, `sam3_labels_extra.jsonl`	SAM 3 boll boxes (analysis views; training-only views)
`...\agrieye_output\objects\boll_cls_roboflow.csv`, `boll_cls_vs_hand.txt`	Roboflow model answer for every boll box; agreement with hand labels
`...\agrieye_output\boll_crops\open\`, `closed\`, `not_boll\`, `unsorted\`, `manifest.csv`	Hand-sorted boll crops
`...\agrieye_output\pov_dataset\`	Training views of all 35 photos, `catalog.csv`, `split.csv`, `leaves.jsonl`, `check_projection.jpg`
`...\agrieye_output\sam3_segmentation\`	SAM 3 leaf outlines of the field
`...\agrieye_output\sam3_yolo_disease\`	SAM 3 leaves judged by disease v2 (`summary.csv`)
`...\agrieye_output\results\detections.csv`, `growth.csv`	**Results per view and per plant + visit**
`...\agrieye_output\results\plots\`	Growth charts (open/closed bolls, leaves, % diseased, canopy, check columns)
`...\agrieye_output\results\povs\`, `station_sheets\`, `leaf_health\`	Annotated views, one sheet per plant, leaf-health pictures
`...\agrieye_output\streetview\tour_2026-07-31.json` … `tour_2026-08-20.json`	Street View tours, one per visit

`F:\Project and completed courses\Agri EYE 360\disease_autolabel\` (leaf and disease work)

Location	Contents
`...\disease_autolabel\manifest.jsonl`	One line per leaf crop: source photo, box, outline, score
`...\disease_autolabel\crops\diseased\`, `healthy\`, `rejected\`	Hand-sorted leaf crops (1,259 / 5,424 / 7,231)
`...\disease_autolabel\dataset_cls\`, `dataset_cls_v3\`, `dataset_leaf\`, `dataset_seg\`	Classifier v2 data, v3 data (field + outside), leaf finder tiles, healthy/diseased segmentation
`...\disease_autolabel\models\disease_cls\weights\best.pt`	**Disease classifier v2 (current)**; v1 backup in `disease_cls_v1_backup\`
`...\disease_autolabel\models\leaf_seg\weights\best.pt`	**Leaf finder**
`...\disease_autolabel\metrics\`	Classifier charts (`00_results_grid.png`, `dashboard.png`, …)

Other result folders

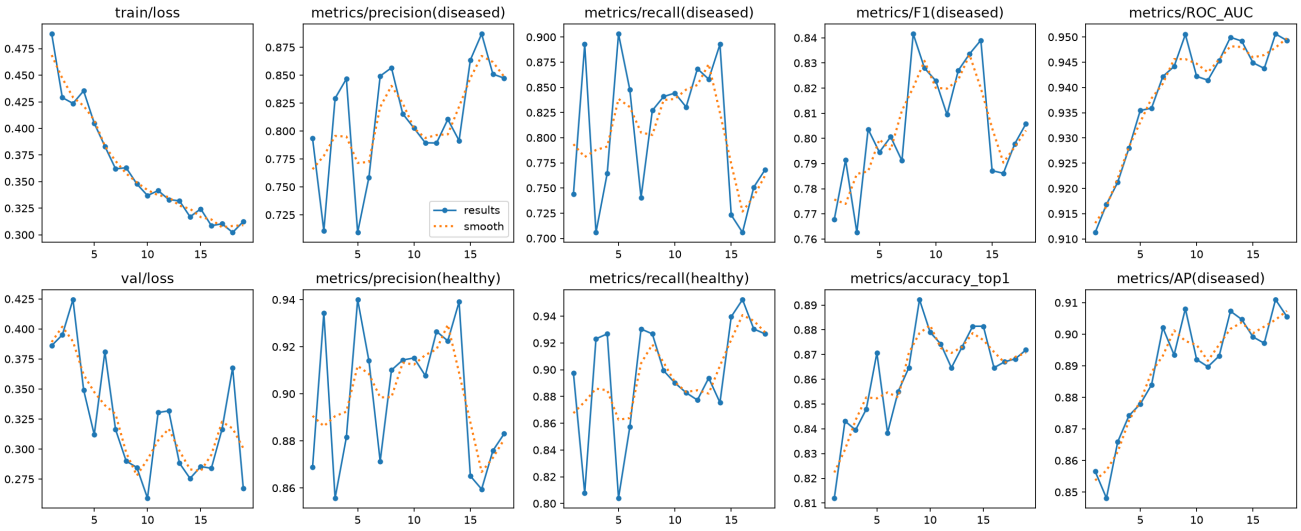
Location	Contents
`F:\Project and completed courses\Agri EYE 360\roboflow_export\`	The 5 Roboflow datasets (folders + zips), `2_sky_negatives_preview.jpg`
`F:\Project and completed courses\Agri EYE 360\sample_results\roboflow_boll_test\`	Roboflow boll model test: `report.txt`, `mistakes.jpg`, `predictions.csv`
`F:\Project and completed courses\Agri EYE 360\sample_results\sam3_seg_check\`	SAM 3 leaf outline check: `report.txt`, `per_view.csv`, `answers.json`
`F:\Project and completed courses\Agri EYE 360\sample_results\boll_figures\`	Open / closed boll pictures (`00_open_vs_closed.jpg`, one file per view)
`F:\Project and completed courses\Agri EYE 360\sample_results\`	Sample outputs, model tests, SAM 3 experiments
`F:\Project and completed courses\Agri EYE 360\Results\`	Workspace links to the segmentation folders and models (junctions: delete with `rmdir`, never `Remove-Item -Recurse`)

4. Environment

- **PC:** Windows 11, Intel i5-1155G7, 16 GB RAM, CPU only (no GPU).
- **Python:** `C:\Users\bhara\Downloads\.venv\Scripts\python.exe` (ultralytics with SAM 3, OpenCV, matplotlib, inference-sdk for Roboflow).
- **Roboflow:** workspace `bharaths-workspace-3fhlw`, free plan (public projects, about 10 credits a month). Keep the private API key out of files, screenshots and GitHub.
- **Before long runs:** set sleep to "never" so the PC stays awake.

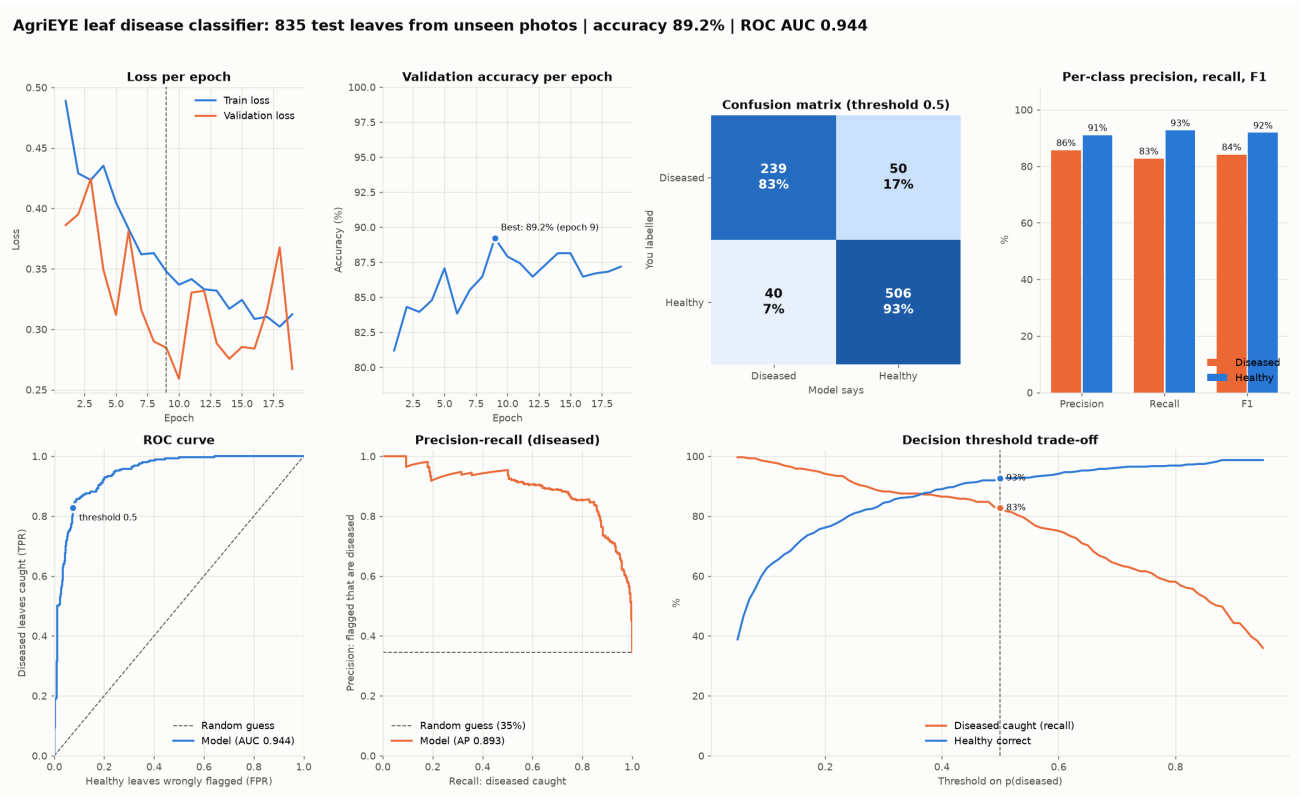
5. Key charts

Disease classifier v2: results per epoch



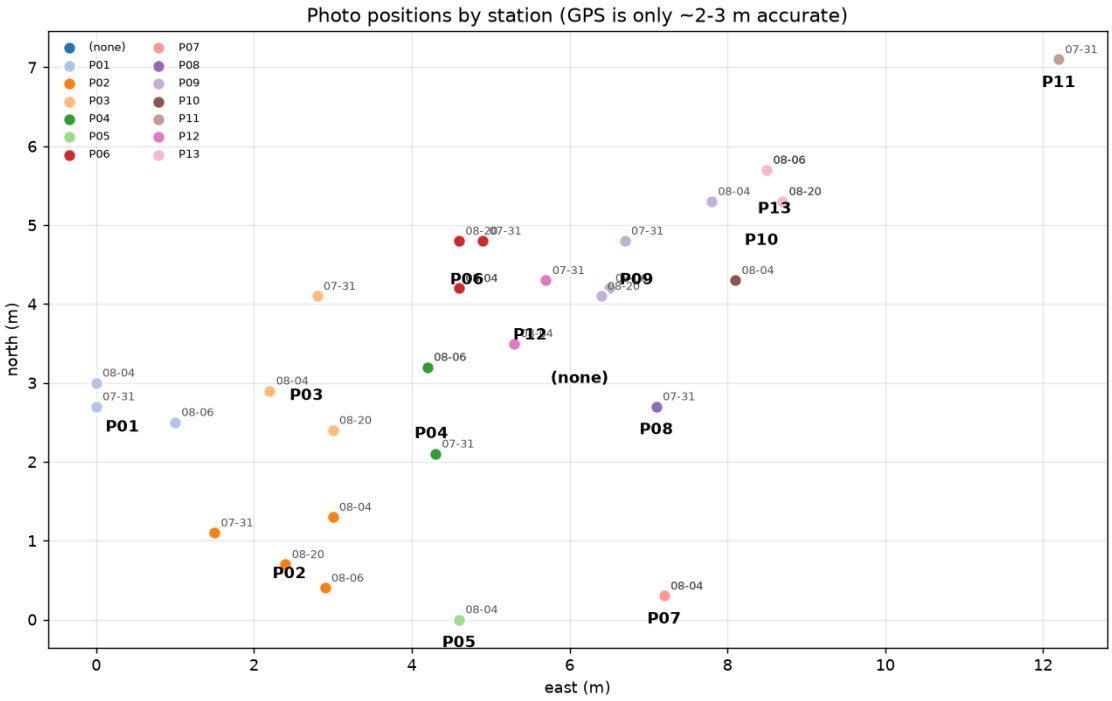
File: F:\Project and completed courses\Agri EYE 360\disease_autolabel\metrics\00_results_grid.png

Disease classifier v2: dashboard



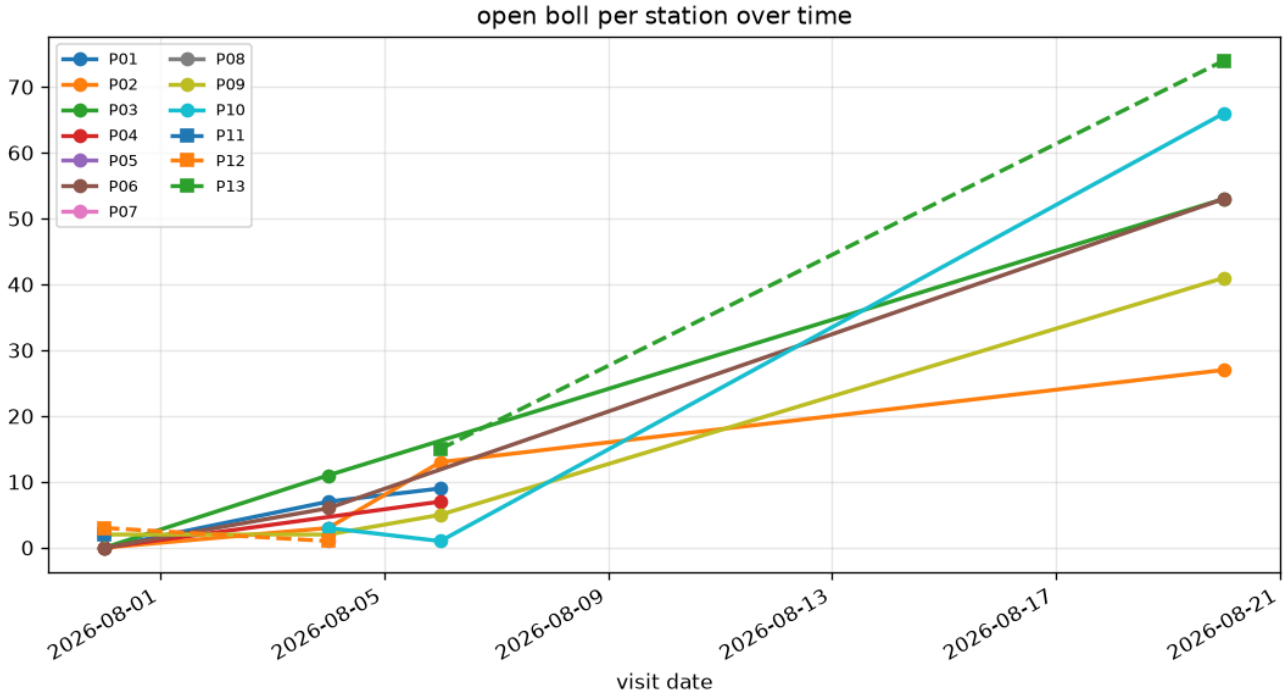
File: F:\Project and completed courses\Agri EYE 360\disease_autolabel\metrics\dashboard.png

Plant (station) positions after the 6 Nov fix



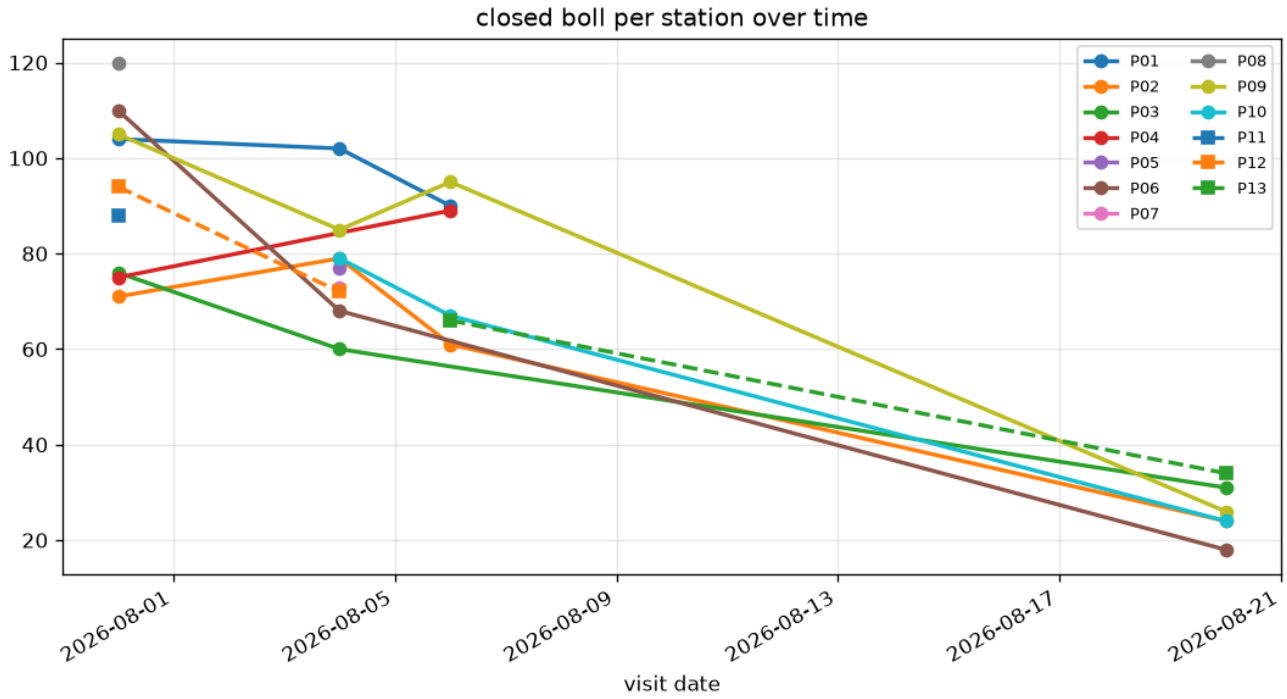
File: F:\Project and completed courses\Agri EYE 360\agrieye_output\station_check\stations_map.png

Open bolls per plant over time (hand labels + Roboflow boll model)



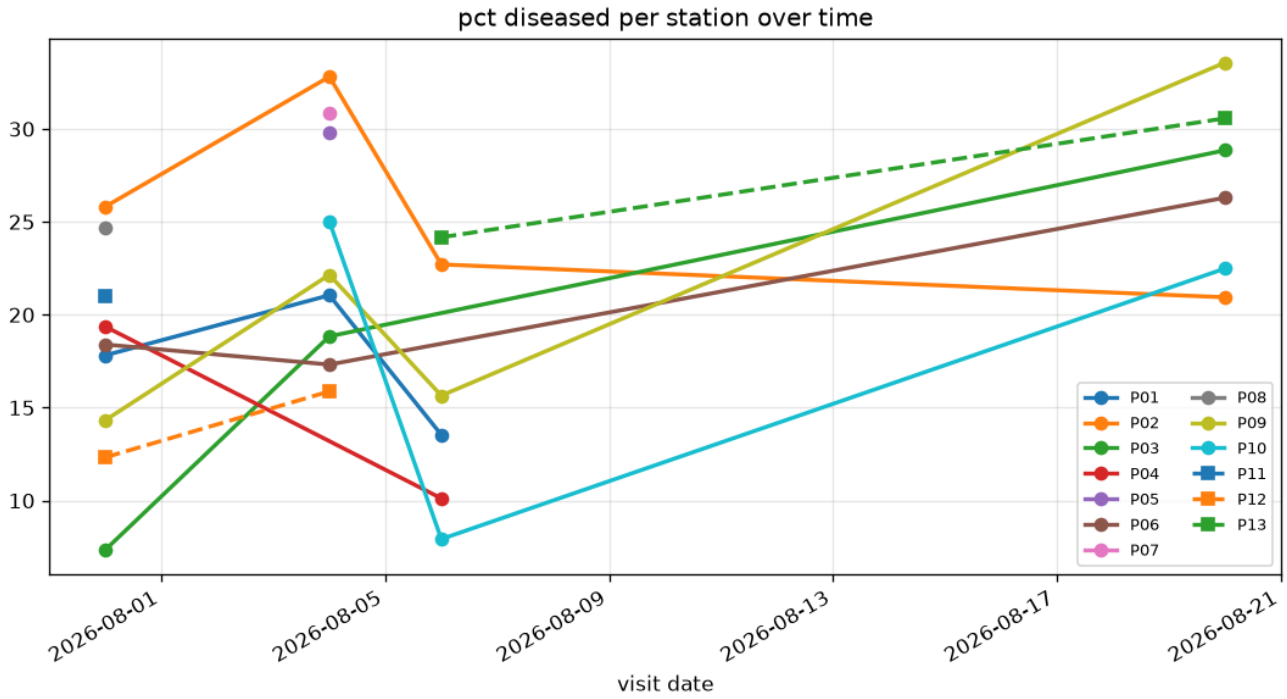
File: F:\Project and completed courses\Agri EYE 360\agrieye_output\results\plots\open_boll.png

Closed bolls per plant over time



File: F:\Project and completed courses\Agri EYE 360\agrieye_output\results\plots\closed_boll.png

% diseased leaves per plant over time



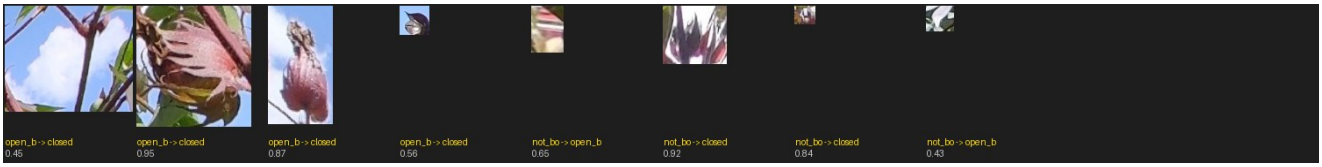
File: F:\Project and completed courses\Agri EYE 360\agrieye_output\results\plots\pct_diseased.png

Hand labels moved from the panorama onto the 4-POV views (check sheet)



File: F:\Project and completed courses\Agri EYE 360\agrieye_output\pov_dataset\check_projection.jpg

Roboflow boll model: every mistake on the 355 test images



File: F:\Project and completed courses\Agri EYE 360\sample_results\roboflow_boll_test\mistakes.jpg

Open (cyan, solid) and closed (magenta, dashed) bolls on the 4 views with the most open bolls



File: F:\Project and completed courses\Agri EYE 360\sample_results\boll_figures\00_open_vs_closed.jpg

SAM 3 leaf outlines (P09, 31 Jul, east view): every outline is a real leaf, but some leaves have none



File: F:\Project and completed courses\Agri EYE 360\sample_results\sam3_seg_check\example_P09_0731_E.jpg